Basic SQL Questions

1. **What is SQL?**
   Answer: SQL (Structured Query Language) is a programming language used for managing and querying relational databases.
2. **What are the different types of SQL commands?**
   Answer:
   - **DDL (Data Definition Language):** CREATE, ALTER, DROP, TRUNCATE
   - **DML (Data Manipulation Language):** INSERT, UPDATE, DELETE
   - **DQL (Data Query Language):** SELECT
   - **DCL (Data Control Language):** GRANT, REVOKE
   - **TCL (Transaction Control Language):** COMMIT, ROLLBACK, SAVEPOINT
3. **What is the difference between SQL and MySQL?**
   Answer: SQL is a query language, whereas MySQL is a database management system that uses SQL.
4. **What is a Primary Key?**
   Answer: A primary key uniquely identifies each record in a table and does not allow NULL values.
5. **What is a Foreign Key?**
   Answer: A foreign key is a column that establishes a relationship between two tables by referencing the primary key of another table.

SQL Joins and Relationships

6. **What are the different types of SQL joins?**
   Answer:
   - **INNER JOIN**: Returns matching rows from both tables.
   - **LEFT JOIN**: Returns all rows from the left table and matching rows from the right.
   - **RIGHT JOIN**: Returns all rows from the right table and matching rows from the left.
   - **FULL JOIN**: Returns all rows when there is a match in either table.
   - **CROSS JOIN**: Returns the Cartesian product of two tables.
   - **SELF JOIN**: A table is joined with itself.
7. **Write a query to fetch employee names and their department names using a join.**
8. SELECT e.name, d.department_name
9. FROM employees e
10. INNER JOIN departments d ON e.department_id = d.id;

11. **What is the difference between WHERE and HAVING?**
    Answer: WHERE filters rows before aggregation, while HAVING filters groups after aggregation.
12. **What is the difference between UNION and UNION ALL?**
    Answer: UNION removes duplicates, whereas UNION ALL includes all rows, including duplicates.
13. **What is a Self-Join? Provide an example.**

```
SELECT e1.name AS Employee, e2.name AS Manager
FROM employees e1
JOIN employees e2 ON e1.manager_id = e2.id;
```

SQL Queries for Data Analysis

11. **Write an SQL query to get the second-highest salary from an Employee table.**

```
SELECT MAX(salary)
FROM employees
WHERE salary < (SELECT MAX(salary) FROM employees);
```

12. **How do you find duplicate records in a table?**

```
SELECT column_name, COUNT(*)
FROM table_name
GROUP BY column_name
HAVING COUNT(*) > 1;
```

13. **How to delete duplicate rows but keep one?**

```
DELETE FROM employees
WHERE id NOT IN (
  SELECT MIN(id) FROM employees GROUP BY name, salary
);
```

14. **What is the difference between RANK(), DENSE_RANK(), and ROW_NUMBER()?**
    Answer:

- RANK(): Assigns ranks with gaps.
- DENSE_RANK(): Assigns consecutive ranks without gaps.
- ROW_NUMBER(): Assigns unique row numbers.

15. **How do you find the top 3 highest salaries?**

```
SELECT DISTINCT salary
```

```
FROM employees
ORDER BY salary DESC
LIMIT 3;
```

Advanced SQL Questions

16. **What are Window Functions in SQL?**
    Answer: Window functions perform calculations across a set of table rows related to the current row. Example: RANK(), LEAD(), LAG().
17. **What is the difference between DELETE, TRUNCATE, and DROP?**
    Answer:

- DELETE: Removes specific rows with a WHERE clause.
- TRUNCATE: Deletes all rows but keeps the table structure.
- DROP: Removes the table completely.

18. **Explain the use of the CASE statement in SQL.**

```
SELECT name, salary,
CASE
  WHEN salary > 50000 THEN 'High'
  WHEN salary BETWEEN 30000 AND 50000 THEN 'Medium'
  ELSE 'Low'
END AS salary_category
FROM employees;
```

19. **What is a Common Table Expression (CTE)?**

```
WITH Employee_CTE AS (
  SELECT id, name, salary FROM employees WHERE salary > 50000
)
SELECT * FROM Employee_CTE;
```

20. **How does indexing work in SQL?**
    Answer: Indexing improves query performance by creating a lookup structure for faster data retrieval.

SQL Performance Optimization

21. **How do you optimize SQL queries?**

- Use indexing.

- Avoid SELECT *.
- Use EXPLAIN to analyze queries.
- Normalize tables.

22. **What is the difference between Clustered and Non-Clustered Index?**
    Answer:

- **Clustered Index**: Sorts the actual data. Only one per table.
- **Non-Clustered Index**: Creates a separate structure pointing to data.

23. **What is a Subquery?**
    Answer: A subquery is a query inside another query.
24. **Write a query to find employees earning more than their department's average salary.**

SELECT name, salary FROM employees e

WHERE salary > (SELECT AVG(salary) FROM employees WHERE department_id = e.department_id);

25. **What is the difference between EXISTS and IN?**
    Answer:

- EXISTS: Checks for existence in a subquery.
- IN: Checks for specific values in a subquery result.

Data Manipulation and Transactions

26. **What is ACID in SQL?**
    Answer:

- **Atomicity**: Transactions are all-or-nothing.
- **Consistency**: Ensures database validity.
- **Isolation**: Transactions are independent.
- **Durability**: Ensures persistence.

27. **How to use transactions in SQL?**

BEGIN TRANSACTION;

UPDATE accounts SET balance = balance - 100 WHERE id = 1;

UPDATE accounts SET balance = balance + 100 WHERE id = 2;

COMMIT;

28. **What is the difference between COMMIT and ROLLBACK?**
    Answer: COMMIT saves changes, whereas ROLLBACK undoes changes.
29. **What is a Deadlock in SQL?**
    Answer: A deadlock occurs when two transactions block each other, preventing execution.

30. **What is a Stored Procedure?**

```
CREATE PROCEDURE GetEmployeeDetails()
AS
BEGIN
  SELECT * FROM employees;
END;
```